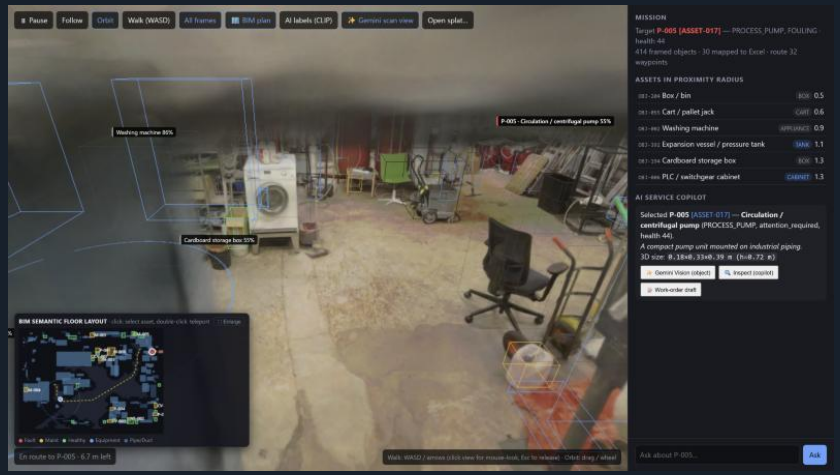
# Semantic digital twin and field-service copilot

theMOLFARS · Valmet × Since AI 2026 · Industrial Metaverse challenge




**414**

3D-framed objects, no overlaps

**30 / 30**

Excel assets mapped

**8.9 m**

collision-free A* route

**49 / 50**

vision benchmark crops

# From blind scan to navigable twin

1

### Sanitize

1.68 M splats read; corrupted tails cut

2

### Floor + grid

Floor/ceiling peaks, 10 cm occupancy grid

3

### Segment

Colour-aware voxel graph + 3D NMS

4

### Recognise

CLIP + video photo match + Gemini

5

### Map Excel

Fit score: class, size, vision

6

### Route

A* with clearance → scene.json

### pipeline/ (Python)

build_scene.py runs offline in ~5 s and writes scene.json, the clean splat and a floor plan

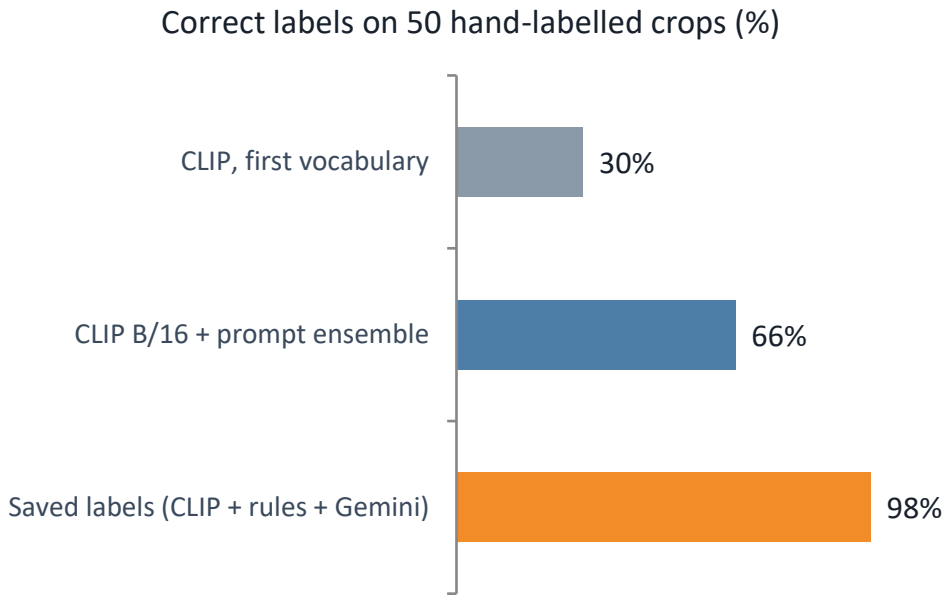
### web/ (three.js + Spark)

Renders the splat, frames, avatar, proximity list and copilot chat; CLIP runs on WebGPU

### server.py (FastAPI)

PostgreSQL evidence, Gemini answers, vision labels, work-order drafts

# 414 non-overlapping, named 3D frames




*49/50 is an upper bound for this room: part of the rules were written for this scan.*

**Colour-aware segmentation**

4 cm voxels join only if their colours match, so a red pump splits from its white pipe

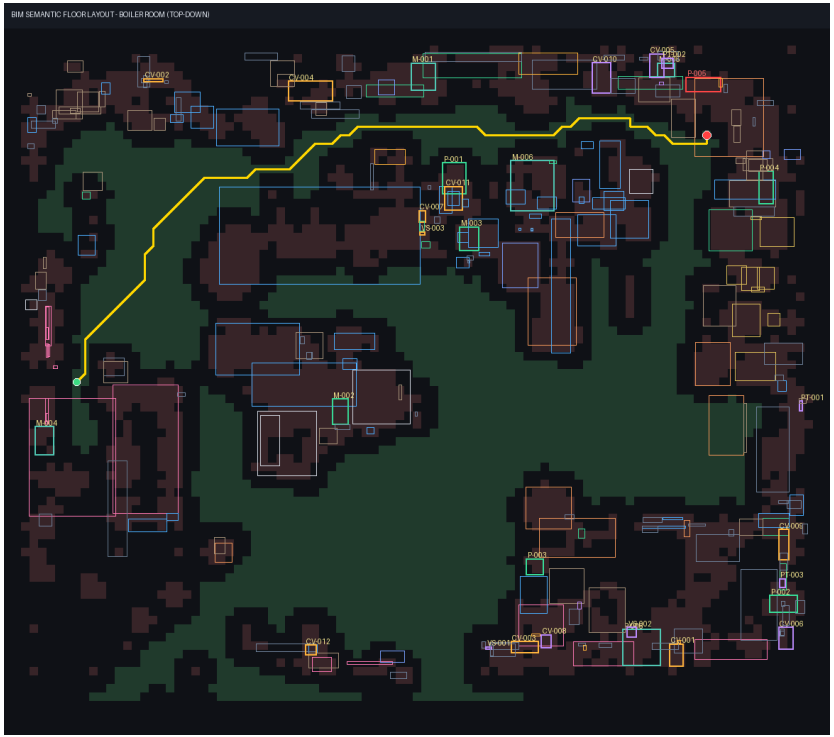
**No stacked frames**

3D NMS drops a box when > 35% of it sits inside a bigger one

**17 classes, honest names**

Pumps, valves, gauges, motors — and the washing machine, boxes and ladder

# A route that never clips through equipment




### Occupancy grid, 10 cm

Floor = splats within ±12 cm; obstacles 0.2–1.7 m high; every cell 0.22 m from the nearest obstacle splat

### A* with clearance cost

Keeps to the middle of aisles; line-of-sight smoothing; every cell asserted walkable

### Proximity radius 1.3 m

Frames and Excel status appear as the avatar passes: 14 assets plus the target on this route

### Walk mode collides

First-person WASD is tested against the same grid, so you slide along walls

# Excel data lands on boxes that fit

### Worst health is the target

ASSET-017 / P-005, process pump, fouling, health 44, on box OBJ-115 that vision reads as a pump

### Fit score per asset

Exact class +4, compatible class +1.5, seen by vision +2, size check per class

### Never on walls or clutter

Boxes read as wall, pipe, box, chair or appliance are excluded; a test fails if that breaks

**Some of the 14 assets the avatar passes within 1.3 m**

| Tag | Class | State | Health |
|---|---|---|---|
| **P-005** | Process pump | attention required | 44 |
| M-006 | Electric motor | attention required | 45 |
| P-001 | Process pump | degraded | 69 |
| M-001 | Electric motor | watch | 77 |
| CV-010 | Control valve | healthy | 96 |
| PT-002 | Pressure transmitter | healthy | 96 |

*Statuses are the Excel values as they are: the route also passes degraded and watch-listed devices.*

# Cited diagnosis, honest confidence



**Arrival at P-005**



**SQL over 8 Excel tables**



**Gemini or offline template**



**4-section cited report**

- What is wrong, why, safe next steps (lockout first), data trust
- Every line cites [EVT-…] [WO-…] [CASE-…] [DOC-…] [DQ-…] (Q-03)
- Follow-up questions in chat; a work-order draft, never submitted (Q-05)
- PostgreSQL in Docker, SQLite from the same Excel when Docker is off

**Confidence for P-005 (Q-06)**

**0.91** model confidence from the pack

**0.43** after 9 open data-quality flags

**0.55** 3D identity of the mapped box

**0.43** overall: the weakest link

# Modular backend that is safe to demo

### Pipeline

- build_scene.py: splat → scene.json
- extract_frames.py: video frames
- gemini_vision.py: batch labels
- video_lift.py: COLMAP + CLIP-L + OWLv2 on GPU

### API (pack's API_Contract)

- /api/assets, /{id}, /telemetry, /events, /documents
- /api/scene/anchors with real boxes
- /api/copilot/query, /api/work-orders/draft
- /api/vision, Gemini object + view scans

### Trust and reliability

- JSON-only POSTs block cross-site writes
- Labels escaped; DB bound to localhost
- Gemini off the event loop; atomic writes
- 13 unit tests; splat cached (35 MB)

# Five-minute demo, and what comes next

1. Orbit overview: 414 frames over the splat
2. Floor plan: grid, obstacles, A* route
3. Start mission: frames light up within 1.3 m
4. Arrival at P-005: cited diagnosis + work-order draft
5. Walk mode: walls stop you

## Next

- Live OPC UA / MQTT feeds behind the same SQL (Q-07)
- Run video_lift.py on the GPU PC for multi-view labels
- OpenUSD export of frames and anchors (Q-09)
- Re-run Gemini labels without the old score floor

**github.com/huziichuk/theMOLFARS**